



KNOX GRAMMAR SCHOOL
MATHEMATICS FACULTY

Set By: IM

Teachers:

DS
IM
IB (Year 12)
IB (Year 11)
RM (Year 11)
SH
LS

2005

TRIAL HSC EXAMINATION

Mathematics (Year 12)

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks (120)

- Attempt Questions 1–10
- All questions are of equal value
- Use a **SEPARATE** Writing Booklet for each question
- Write your **Board of Studies Student Number and Class Teacher's Initials** on the front cover of each of your writing booklets

Board of Studies Student Number: _____

Class Teacher's Initials: _____

(Year 12)
(Year 11)
11

(Year 12)
(Year 11)
value

(Year 12)
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(Year 11)
11

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Total marks (120)

Attempt questions 1 – 10

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)

Marks

- (a) Evaluate $\frac{3}{\log_e 9}$ correct to three significant figures. **2**
- (b) Differentiate with respect to x the expression $\frac{1}{2x} + \pi$. **2**
- (c) Simplify $\frac{2x}{3} - \frac{x+2}{5}$. **2**
- (d) Find a primitive of $2 - \sqrt{x}$. **2**
- (e) Solve $1 - 9x^2 = 0$. **2**
- (f) Peter paid \$1.15 per litre of petrol. This was 12.5% above the recommended retail price. Calculate the recommended retail price per litre of petrol. **2**

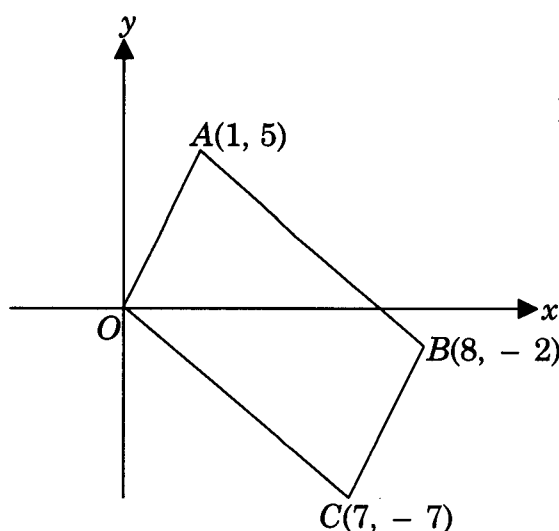


Diagram is not to scale

In the diagram, $OABC$ is a quadrilateral where the coordinates of O , A , B and C are $(0, 0)$, $(1, 5)$, $(8, -2)$ and $(7, -7)$ respectively.

- | | | |
|-----|---|---|
| (a) | Find the midpoint of the interval joining AC . | 1 |
| (b) | Find the slope of AB . | 1 |
| (c) | Show that the equation of the line containing AB is $x + y = 6$. | 1 |
| (d) | Find the exact length of AB . | 2 |
| (e) | Show that OC is parallel to AB . | 1 |
| (f) | Explain why $OABC$ is a parallelogram. | 2 |
| (g) | Find the perpendicular distance from O to AB . Leave your answer in simplest surd form. | 2 |
| (h) | Find the area of parallelogram $OABC$. | 2 |

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Find:

(i) $\frac{d}{dx}(x^2 e^{3x})$.

2

(ii) $\frac{d}{dx}\left(\frac{\sin x}{x}\right)$.

2

(b)

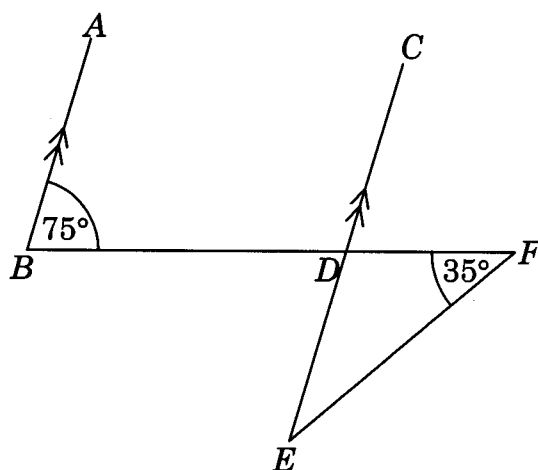


Diagram is not to scale

In the diagram, $AB \parallel CE$, $\angle ABF = 75^\circ$ and $\angle BFE = 35^\circ$.

2

Find the size of $\angle DEF$, giving reasons.

(c) The graph of $y = f(x)$ passes through the point $(2, 12)$ and $f'(x) = 9x^2 + 4$.

2

Find $f(x)$.

(d) Find $\int \frac{4x}{x^2 - 3} dx$.

2

(e) Find $\int_{\frac{\pi}{6}}^{\pi} \cos 2x \, dx$.

2

- (a) The equation of a parabola is $(x+3)^2 = -12(y-1)$.
- (i) Write down the coordinates of the vertex of this parabola. 1
- (ii) State the coordinates of the focus. 1
- (iii) State the equation of the directrix. 1
- (iv) Sketch the graph of the parabola, showing the above features. 2
- (b) (i) Express $0.\dot{2}\dot{5}$ as an infinite geometric series. 1
- (ii) Hence, write $0.\dot{2}\dot{5}$ as a rational number in lowest terms. 1

(c)

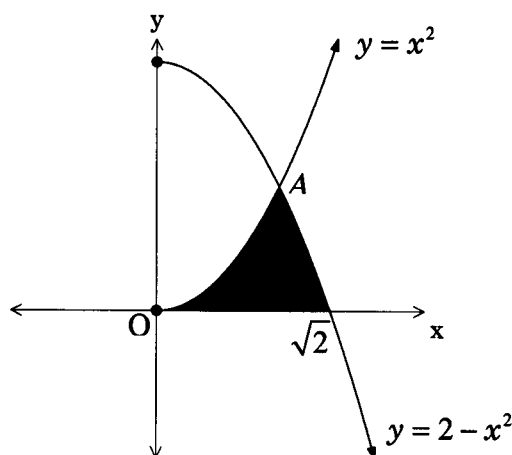


Diagram is not to scale

The diagram above shows the graphs of $y = x^2$ and $y = 2 - x^2$ intersecting at the point A. The curve $y = 2 - x^2$ crosses the x-axis at $(\sqrt{2}, 0)$.

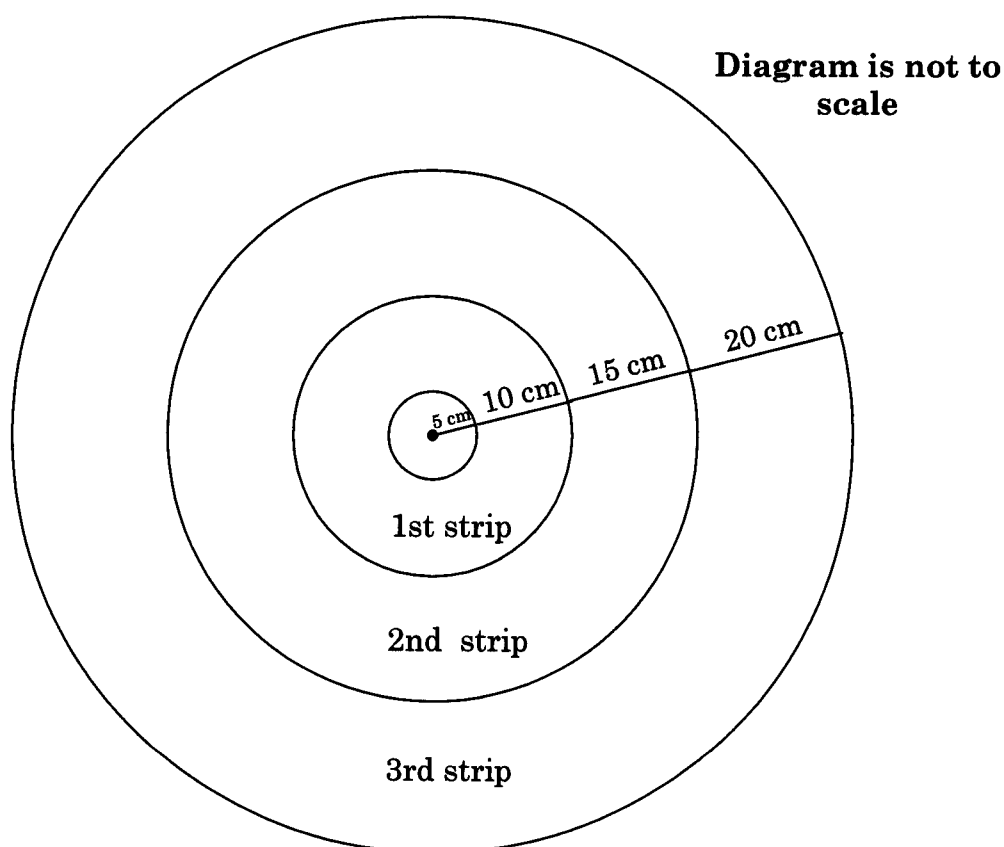
Note that $x \geq 0$.

- (i) Find the x-coordinate of A. 1
- (ii) Find the area of the shaded region bounded by $y = x^2$, $y = 2 - x^2$ and the x-axis. Give your answer correct to two decimal places. 4

(a) Consider the function $f(x) = 2x^3 - 6x^2 - 18x + 1$.

- | | | |
|-------|---|---|
| (i) | Find the coordinates of the stationary points on the curve $y = f(x)$ and determine their nature. | 4 |
| (ii) | Find the values of x for which $y = f(x)$ is increasing. | 2 |
| (iii) | For what value of x is the rate of decrease the greatest. | 1 |

(b)



Beginning with a circular piece of fabric of radius 5 cm, Susan sewed together circular strips of different coloured fabrics which increased in width to make a circular table cloth as shown in the diagram. The finished width of the first strip was 10 cm and of the second strip was 15 cm and so on.

- | | | |
|------|--|---|
| (i) | Show that the width of the 10 th strip was 55 cm. | 2 |
| (ii) | How many strips must be sewn together to complete a circular table cloth of radius 455 cm? | 3 |

- (a) Consider the equation $\log_e y = x \log_e \left(\frac{1}{2} \right)$, where $y > 0$.

- (i) Write down an expression for $y = f(x)$. 2
- (ii) Sketch the graph of $y = f(x)$ and label the axes appropriately. 1

(b)

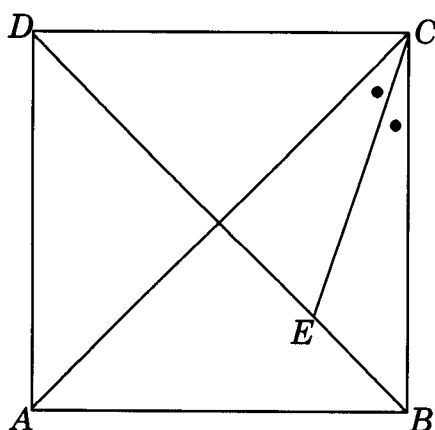


Diagram is not to scale

In the diagram, $ABCD$ is a square with diagonals AC and BD . The point E lies on DB and the interval CE bisects $\angle ACB$.

Copy or trace this diagram into your writing booklet.

- (i) Prove, with reasons, that $\angle DCE = \angle DEC$. 2
- (ii) Hence, show that $DE = DA$. 2

- (c) An object falls into a tank filled with oil. The rate of decrease of the velocity is proportional to its velocity v centimetres per second. This statement can be expressed mathematically by the equation $\frac{dv}{dt} = -kv$, where k is a constant and t is time measured in seconds.

The initial speed of the object when it enters the tank is 85 cm/s. Five seconds later, the speed of the object is 60 cm/s.

- (i) Verify that $v = Ae^{-kt}$, where A is a constant, satisfies $\frac{dv}{dt} = -kv$. 1
- (ii) Find the values of A and k . 3
- (iii) Calculate the velocity of the object when $t = 8$. 1

(a)

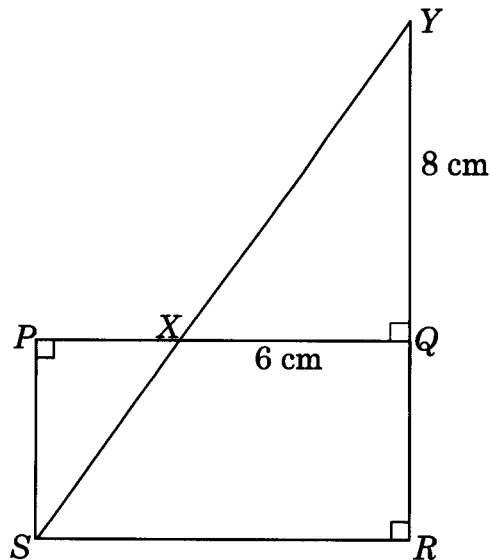


Diagram is not to scale

In the diagram, $PQRS$ is a rectangle and $SR = 3PS$. The points R , Q and Y are collinear points. $XQ = 6$ cm and $YQ = 8$ cm.

(i) Prove that $\triangle PXS \parallel \triangle QXY$.

2

(ii) Find the length of PS .

2

(b) The velocity of a particle moving in a straight line is given by $v = 1 - 2\cos t$ for $0 \leq t \leq 2\pi$, where v is measured in metres per second and time t is measured in seconds.

(i) At what times in the interval $0 \leq t \leq 2\pi$, is the particle at rest?

2

(ii) Sketch the graph of v against t for $0 \leq t \leq 2\pi$.

2

(iii) What is the maximum velocity of the particle in the interval $0 \leq t \leq 2\pi$?

1

(iv) Calculate the total distance travelled by the particle in the first π seconds.

3

Question 8 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Consider the function $y = \ln(x - 2)$ for $x > 2$.

(i) Sketch the graph of $y = \ln(x - 2)$, showing essential features.

2

(ii) Use Simpson's rule with three function values to evaluate:

2

$$\int_3^5 \ln(x - 2) \, dx$$

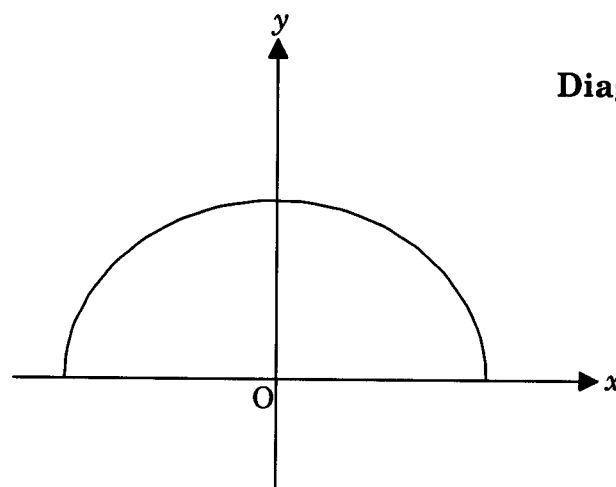
Round off your answer to one decimal place.

(b) The quadratic equation $x^2 + mx + n = 0$ has one root that is twice the other. Let one of the roots be α .

4

Using the equations for the sum and product of the roots of $x^2 + mx + n = 0$, find the value of $\frac{m^2}{n}$.

(c)



The diagram shows part of the curve with equation $\frac{x^2}{9} + \frac{y^2}{4} = 1$ which lies above the x -axis.

(i) Find the x -intercepts.

1

(ii) Find the volume of the solid of revolution formed when this curve is rotated about the x -axis. Leave your answer in simplified form in terms of π .

3

- (a) Two cards are chosen at random from the four cards shown below.



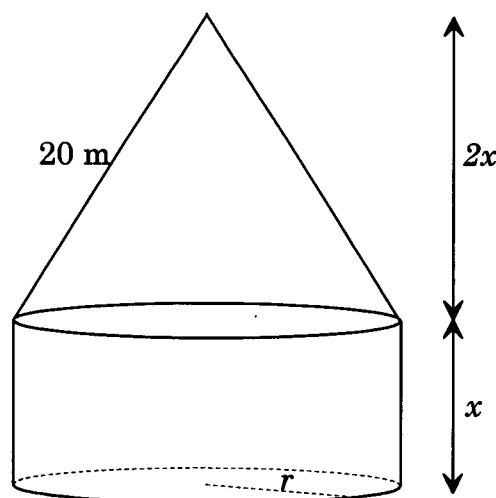
A person holds the two chosen cards in their hand.

- | | | |
|-------|---|---|
| (i) | Calculate the probability that the sum of the numbers on the cards chosen is zero. | 1 |
| (ii) | Find the probability that at least one of the cards in the person's hand is a 2. | 1 |
|
 | | |
| (b) | A water tank which is initially full is being emptied. The remaining quantity of water, L litres, in the tank, at any time, t minutes, after it starts to empty is given by $L = 1000(20 - t)^2$ for $0 \leq t \leq 20$. | |
| (i) | How much water was in the tank initially? | 1 |
| (ii) | Show that $\frac{dL}{dt} = 2000t - 40\,000$. | 1 |
| (iii) | At what time is the water flowing out of the tank at a rate of 20 000 litres per minute? | 1 |
| (iv) | How long will it take to empty the tank? | 1 |
| (iv) | At what rate is the tank emptying when the tank is half full? | 2 |
|
 | | |
| (c) | Consider the function $f(x) = \pi x - \cos \pi x$. | |
| (i) | Find $f''\left(\frac{1}{2}\right)$. | 2 |
| (ii) | Show that the point $\left(\frac{1}{2}, \frac{\pi}{2}\right)$ is a point of inflexion on the graph of $y = f(x)$. | 2 |

Question 10 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) An initial sum of \$2000 is invested into a retirement fund and interest is compounded at the rate of 8% per annum, every six months.
- (i) If no further deposits are made into this fund, calculate how much the initial investment is worth at the end of 20 years? **2**
- (ii) Now suppose that at the beginning of the second year, and the beginning of each subsequent year, a further \$500 is deposited into the fund. Determine how much the fund is worth at the end of 20 years? **4**

(b)



A grain silo has a cylindrical shaped wall and a cone shaped roof as shown in the diagram. Let the radius of the base of the silo be r metres and the height of the cylinder be x metres. The height of the cone is $2x$ metres.

- (i) Show that if the length of the slant edge of the cone is 20 metres, then $r^2 = 400 - 4x^2$. **1**
- (ii) Show that the volume, V , in cubic metres, of the silo is given by: **2**

$$V = \frac{20\pi}{3}(100x - x^3)$$

- (iii) Find the exact height of the silo so that it holds a maximum amount of grain. **3**

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note $\ln x = \log_e x, \quad x > 0$

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QUESTION 1 (DS)

a) $\frac{3}{\log_e 4} = 1.3653588 \dots$ ①

$= 1.37$ (3 S.F.) ①

b) $\frac{d}{dx} \left(\frac{1}{2x} + \pi \right) = \frac{d}{dx} \left(\frac{1}{2} x^{-1} + \pi \right)$
 $= -\frac{1}{2} x^{-2}$
 $= -\frac{1}{2x^2}$ ②

c) $\frac{2x}{3} - \frac{x+2}{5} = \frac{10x - 3(x+2)}{15}$
 $= \frac{7x-6}{15}$ ②

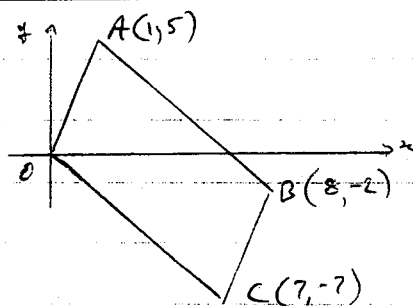
d) primitive of $2 - x^{1/2} = 2x - \frac{2}{3} x^{3/2}$ ②

e) $1 - 9x^2 = 0$
 $(1 - 3x)(1 + 3x) = 0$
 $x = \pm \frac{1}{3}$ ②

f) $112\frac{1}{2}\% = 115$
 $1\% = 1.0222 \dots$
 $100\% = 102.22 \dots$

Recommended retail
 price is 102.2 cents/litre.
 ②

QUESTION 2 (SH)



a) Midpoint AC = (4, -1) ①

b) Slope of AB = -1 ①

c) Equ. AB,
 $y - 5 = -1(x - 1)$
 $y - 5 = -x + 1$
 $x + y = 6$ ①

d) length AB = $\sqrt{(8-1)^2 + (-2-5)^2}$ ①
 $= \sqrt{98}$
 $= 7\sqrt{2}$ units ①

e) Slope of OC = $\frac{-7-0}{7-0}$
 $= -1$ ①
 $\therefore AB \parallel OC$

f) Slope OA = 5, Slope BC = 5 ①
 Since OA $\parallel$ BC and OC $\parallel$ AB (from above)
 then opposite sides are parallel ①
 and OABC is a parallelogram.

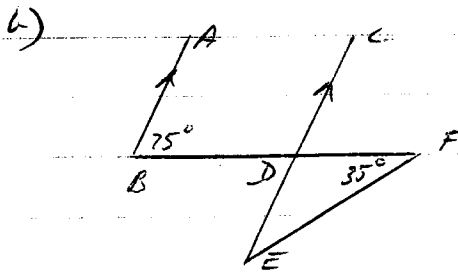
g) $\perp d = \frac{|1 \times 0 + 1 \times 0 + 6|}{\sqrt{1^2 + 1^2}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ units ①

h) Area OABC = $7\sqrt{2} \times 3\sqrt{2}$ ①
 $= 42$ units² ①

QUESTION 3 (AM)

a) i) $\frac{d}{dx} x^2 e^{3x} = e^{3x} \times 2x + x^2 \times 3e^{3x}$
 $= 2xe^{3x} + 3x^2 e^{3x}$ (2)
 $= xe^{3x}(2 + 3x)$

ii) $\frac{d}{dx} \frac{\sin x}{x} = \frac{x \times \cos x - \sin x \times 1}{x^2}$
 $= \frac{x \cos x - \sin x}{x^2}$ (2)



$\angle CDF = 75^\circ$ Corresponding angles.
 $\angle DEF + 35^\circ = 70^\circ$ Exterior angle $\triangle DEF$
 $\therefore \angle DEF = 40^\circ$ (2)

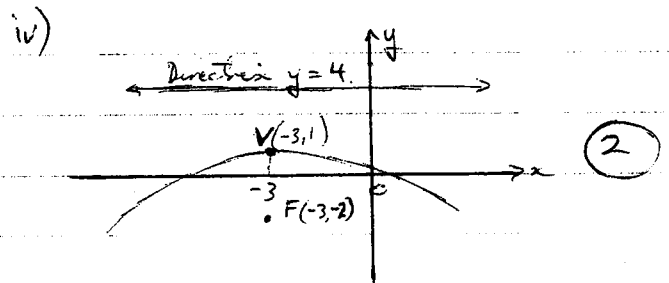
c) $f'(x) = 8x^2 + 4$
 $f(x) = 3x^3 + 4x + C$
 $f(x) = 12$ when $x = 2$
 $\therefore 12 = 24 + 8 + C \Rightarrow C = -20$
 $\therefore f(x) = 3x^3 + 4x - 20$ (2)

d) $\int \frac{4x}{x^2 - 3} dx = 2 \log_e(x^2 - 3) + C$ (1)

e) $\int_{\pi/6}^{\pi} \cos 2x dx = \left[\frac{1}{2} \sin 2x \right]_{\pi/6}^{\pi}$ (1)
 $= \frac{1}{2} \left(0 - \frac{\sqrt{3}}{2} \right)$
 $= -\frac{\sqrt{3}}{4}$ (1)

QUESTION 4 (IB)

a) $(x+3)^2 = -12(y-1)$
 i) Vertex = $(-3, 1)$ (1)
 ii) Focus = $(-3, -2)$ (1)
 iii) Directrix is $y = 4$ (1)

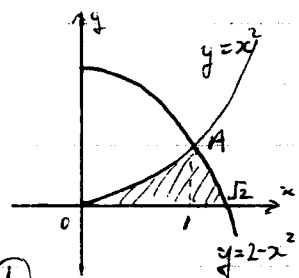


b) i) $0.\dot{2}\dot{5} = \frac{25}{100} + \frac{25}{10000} + \frac{25}{1000000} + \dots$ (1)

ii) Using $S_{\infty} = \frac{a}{1-r}$,
 $0.\dot{2}\dot{5} = \frac{25}{100} \times \frac{100}{99}$
 $= \frac{25}{99}$ (1)

c) i) $x^2 = 2 - x^2$
 $2x^2 - 2 = 0$
 $x^2 - 1 = 0$
 $x = \pm 1$

$\therefore$ x-coordinate of A is 1. (1)



ii) Area = $\int_0^1 x^2 dx + \int_1^{\sqrt{2}} 2 - x^2 dx$ (1)

$= \left[\frac{x^3}{3} \right]_0^1 + \left[2x - \frac{x^3}{3} \right]_1^{\sqrt{2}}$ (1)

$= \left(\frac{1}{3} - 0 \right) + \left[\left(2\sqrt{2} - \frac{2\sqrt{2}}{3} \right) - \left(2 - \frac{1}{3} \right) \right]$

$= \frac{4\sqrt{2}}{3} - \frac{4}{3}$ (1)

$= 0.55$ (2dp) (1)

QUESTION 6 CONT'D.

c ii) $v = Ae^{-kt}$

$t=0, v=85$

$\therefore 85 = Ae^0$

$\therefore A = 85$ (1)

$v = 85e^{-kt}$

$t=5, v=60$

$\therefore 60 = 85e^{-5k}$

$\frac{60}{85} = e^{-5k}$

$\ln\left(\frac{60}{85}\right) = -5k$

$k \doteq 0.0696613 \dots$ (2)

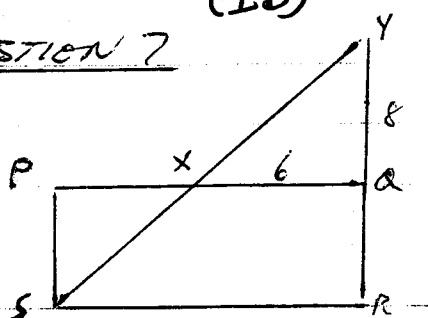
iii) When $t=8$

$v = 85e^{-8k}$

$v \doteq 48.68 \text{ cm/s}$ (1)

QUESTION 7

a)



In $\triangle PXS$ & $\triangle QXY$.

i) $\angle PXS = \angle QXY$ (Vertically opposite)

$\angle SPX = \angle XQR$ (right angles of rect. PQRS)

$= \angle YQX$ (straight angle)

$\therefore \triangle PXS \parallel \triangle QXY$ (2 equal angles)

(2)

ii) $\frac{PS}{YA} = \frac{PX}{QX}$ corresponding sides of similar $\triangle$ s.

$\therefore \frac{x}{8} = \frac{3x-6}{6}, PS=x$

$6x = 24x - 48$ (2) Note: $SR = 3PS$

$18x = 48 \therefore x = \frac{2}{3}, PS = \frac{2}{3}$

b) $v = 1 - 2\cos t \quad 0 \leq t \leq 2\pi$

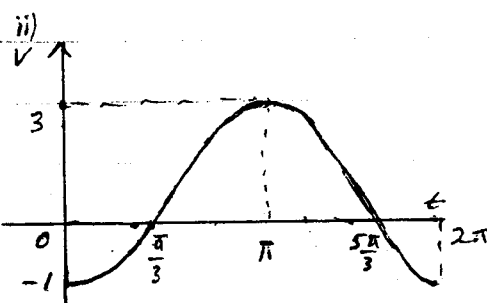
i) At rest, $v=0$.

$1 - 2\cos t = 0$

$\cos t = \frac{1}{2}$

$t = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$ (2)

At rest when $t = \frac{\pi}{3}$ and $\frac{5\pi}{3}$ secs.



(2)

iii) Maximum velocity is 3 m/s (1)

iv) Distance travelled is given by

$\int_0^{\pi/3} -(1-2\cos t) dt + \int_{\pi/3}^{\pi} (1-2\cos t) dt$ (1)

distance = $\int_0^{\pi/3} 2\cos t - 1 dt + \int_{\pi/3}^{\pi} 1 - 2\cos t dt$

$= [2\sin t - t]_0^{\pi/3} + [t - 2\sin t]_{\pi/3}^{\pi}$ (1)

$= (2 \times \frac{\sqrt{3}}{2} - \frac{\pi}{3}) - (0) + (\pi - 0) - (\frac{\pi}{3} - 2 \times \frac{\sqrt{3}}{2})$

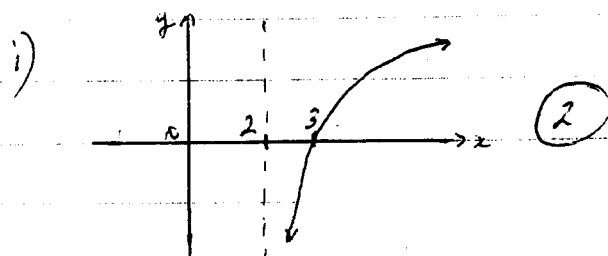
$= \sqrt{3} - \frac{\pi}{3} + \pi - \frac{\pi}{3} + \sqrt{3}$

$= 2\sqrt{3} + \frac{\pi}{3} \text{ metres}$ (1)

QUESTION 8

(IB)

a) $y = \ln(x-2) \quad x > 2$



ii)

$$\int_3^5 \ln(x-2) dx \approx \frac{h}{3} [y_0 + y_2 + 4y_1] \text{ S.R. 3 f.u.}$$

$$= \frac{1}{3} [0 + \ln 3 + 4 \ln 2] \quad (1)$$

$$= 1.3 \text{ (1 d.p.)} \quad (1)$$

b) $x^2 + mx + n = 0$

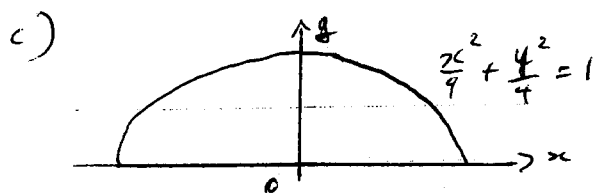
Let the roots be α and 2α

$\therefore$ Sum, $3\alpha = -m \quad (1)$

Product, $2\alpha^2 = n \quad (1)$

$\therefore \frac{m^2}{n} = \frac{9\alpha^2}{2\alpha^2} \quad (1)$

$= \frac{9}{2} \quad (1)$



i) when $y=0$, $\frac{x^2}{9} + 0 = 1$

$$x^2 - 9 = 0$$

$$x = \pm 3 \quad (1)$$

ii) $V = \pi \int_a^b y^2 dx$

Note: $y^2 = 4 - \frac{4x^2}{9}$

$\therefore V = 2\pi \int_0^3 \left(4 - \frac{4x^2}{9}\right) dx \quad (1)$

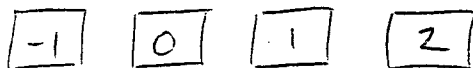
$= 2\pi \left[4x - \frac{4}{27}x^3\right]_0^3 \quad (1)$

$= 2\pi [(12 - 4) - 0]$

$\therefore \text{Volume} = 16\pi \text{ units}^3 \quad (1)$

QUESTION 9 (SH)

a)



i) $P(\text{sum is zero}) = \frac{1}{6} \quad (1)$

ii) $P(\text{one card is 2}) = \frac{1}{2} \quad (1)$

b) $L = 1000(20-t)^2$

i) when $t=0$, $L = 400000$

$\therefore 400000 \text{ L of water initially} \quad (1)$

ii) $L = 1000(20-t)^2$

$\frac{dL}{dt} = 2 \times 1000(20-t)^1 \times -1$

$= -2000(20-t)$

$= -40000 + 2000t$

$= 2000t - 40000 \quad (1)$

QUESTION 9 cont'd.

$$\text{iii)} -20000 = 2000t - 40000$$

$$20000 = 2000t$$

$$t = 10$$

Water is flowing out at 20 000 L/min
when $t=10$ min. (1)

iv) 20 min to empty tank (1)

v) Tank half full,

$$200000 = 1000(20-t)^2$$

$$200 = (20-t)^2$$

$$\pm 10\sqrt{2} = 20-t$$

$$\therefore t = 20 - 10\sqrt{2} \text{ or } (20 + 10\sqrt{2}) \quad 0 \leq t \leq 20$$

$$\therefore t = 20 - 10\sqrt{2} \text{ or } 5.86 \text{ min 2dp}$$

$$\therefore \frac{dL}{dt} = 2000(20 - 10\sqrt{2}) - 40000$$

$$(2) = -20000\sqrt{2} \text{ or } -28284.27$$

$\therefore$ tank is emptying at a rate
of 20 000 $\sqrt{2}$ L/min when half full.

$$\text{c)} f(x) = \pi x - \cos \pi x$$

$$\text{i)} f'(x) = \pi + \pi \sin \pi x$$

$$f''(x) = \pi^2 \cos \pi x$$

$$f''\left(\frac{1}{2}\right) = 0$$

(2)

$$\text{ii)} \text{ when } x = \frac{1}{2}, f(x) = \frac{\pi}{2} - \cos \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\therefore \left(\frac{1}{2}, \frac{\pi}{2}\right) \text{ lies on } f(x) \quad (1)$$

$$f''\left(\frac{1}{2}\right) = 0$$

$$f''\left(\frac{1}{2}-\epsilon\right) > 0 \text{ i.e. } \pi^2 \cos \frac{\pi}{4} > 0$$

$$f''\left(\frac{1}{2}+\epsilon\right) < 0 \text{ i.e. } \pi^2 \cos \frac{3\pi}{4} < 0$$

Since concavity changes about $x = \frac{1}{2}$
then $\left(\frac{1}{2}, \frac{\pi}{2}\right)$ is a pt of inflection.

(1)

QUESTION 10. (IM).

$$\text{a) i)} A = P\left(1 + \frac{r}{100}\right)^n$$

$$= 2000\left(1 + \frac{4}{100}\right)^{40}$$

$$= 2000(1.04)^{40}$$

$$= \$9602.04$$

(1)

(1)

ii)

$$\text{Amount} = 500 \times 1.04^{38} + 500 \times 1.04^{36} + \dots + 500 \times 1.04^2$$

(1)

$$= \frac{500 \times 1.04^2 [(1.04^2)^{19} - 1]}{(1.04^2)^2 - 1}$$

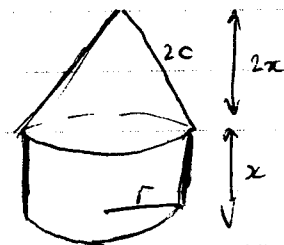
(1)

$$= \$22790.57$$

(1)

$\therefore$ Amount in account at end of
20 years is $\$9602.04 + \22790.57
i.e. $\$32392.61$ (1)

b)



$$\text{i)} r^2 + (20-x)^2 = 20^2$$

$$r^2 + 4x^2 = 400$$

$$\therefore r^2 = 400 - 4x^2$$

(1)

QUESTION 10 CONT'D.

$$\begin{aligned}\text{ii) Volume of silo, } V &= \pi r^2 h + \frac{1}{3} \pi r^2 h \\ &= \pi r^2 x + \frac{1}{3} \pi r^2 (2x) \\ &= \frac{5}{3} \pi r^2 x \\ &= \frac{5}{3} \pi x (400 - 4x^2) \\ &= \frac{20}{3} \pi (100x - x^3) \quad (1)\end{aligned}$$

$$\begin{aligned}\text{iii) } \frac{dV}{dx} &= \frac{20}{3} \pi (100 - 3x^2) \\ \frac{d^2V}{dx^2} &= \frac{20}{3} \pi (-6x) \quad (1) \\ &= -40 \pi x\end{aligned}$$

Stationary pts, $\frac{dV}{dx} = 0$

$$\begin{aligned}\text{ie. } \frac{20\pi}{3} (100 - 3x^2) &= 0 \\ 3x^2 &= 100 \\ x^2 &= \frac{100}{3} \\ x &= \frac{10}{\sqrt{3}} \quad (x > 0) \quad (1)\end{aligned}$$

$$\frac{d^2V}{dx^2} < 0 \text{ for } x = \frac{10}{\sqrt{3}} \quad (1)$$

$\therefore$ Maximum occurs when $x = \frac{10}{\sqrt{3}}$

$$\begin{aligned}\therefore \text{required height} &= 3 \times \frac{10}{\sqrt{3}} \\ &= 10\sqrt{3} \text{ metres.} \\ &\quad (1)\end{aligned}$$